

Hybrid Ensemble Machine Learning Algorithm for Heterogeneous Health Care Data Analytics

Abstract

Accurate classification and detection of medical data are crucial for early disease diagnosis and reducing mortality rates. Machine learning techniques have emerged as powerful tools to assist healthcare professionals by identifying complex patterns in health-related data. However, relying on a single machine learning algorithm often results in performance limitations. To address this, we propose a hybrid ensemble machine learning approach for analyzing heterogeneous healthcare data. The proposed method integrates Logistic Regression, YOLOv4, Parallel K-Means Clustering, and XGBoost to enhance classification accuracy and detection efficiency. Logistic Regression is utilized for initial image classification, while YOLOv4 performs object detection within medical images. Parallel K-Means Clustering aids in efficient data clustering, and XGBoost further refines the output to improve overall accuracy. Experimental results on multiple medical datasets demonstrate that the hybrid approach outperforms individual techniques, delivering superior performance and reliability in healthcare data analysis.

Key words: Logistic regression, YOLOv4, XG boost, clustering, healthcare.

1. Introduction

Healthcare analytics is essential in enhancing patient outcomes, optimizing resource allocation, and facilitating decision-making processes in the era of data-driven management. Classical machine learning methods face substantial challenges due to the diverse composition of healthcare data, which includes structured, unstructured, and semi-structured data from many sources including electronic health records (EHRs), medical imaging, sensor data, and genomic sequences[1]. Effective analysis and extraction of meaningful insights from this diverse data necessitate the use of advanced techniques that surpass single-model solutions. An effective method for addressing these challenges is the Hybrid Ensemble Machine Learning Algorithm, which integrates a number of different machine learning models in order to enhance the accuracy, reliability, and generalization of predictions[2]. Ensemble learning is a method that utilizes the benefits of multiple algorithms in order to overcome the limitations of individual models and adapt to a wide range of healthcare data. In addition to taking into account continuous and categorical variables, this method also takes into account patterns that are present across space and time[3]. This method holds a great deal of promise for the field of healthcare analytics, particularly in areas where precision, accuracy, and dependability are of the utmost importance. The use of hybrid ensemble algorithms allows for the efficient management of various types of data, the handling of missing information, the reduction of biases, and the resolution of class imbalances that are frequently reported in medical datasets[4]. Because of this, they are a potentially useful instrument for a wide variety of applications, such as the prediction of diseases, personalized medicine, the management of population health, and the optimization of treatment. In this work, we will investigate the impact that hybrid ensemble machine learning algorithms have had on a variety of healthcare



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applications by delving into their structure, implementation, and the benefits that they offer for a wide range of healthcare data.

In the opinion of data scientists and machine learning specialists, K-Means clustering is an efficient unsupervised method that can be utilized to discover the structure of a dataset[5]. Over time, the clustering algorithm has established itself as the de facto standard for data partitioning due to the fact that it is both straightforward and efficient. For the purpose of data clustering, this technology has found practical application in a variety of significant real-world contexts, such as cyber security, recommendation systems, and a variety of smart city services[6]. In addition, clustering is an excellent method for providing an evaluation of the data of a company. In addition, the K-means algorithm has been utilized in the context-aware services as well as the user behavior analysis. A significant amount of reliance is placed on the K-means algorithm by feature extraction systems that are complex[7].

Through a process known as machine learning, computers are able to automatically learn new things from data. In contrast to rule-based systems, machine learning is based on the utilization of complex algorithms and statistical methods to perform data analysis and generate accurate predictions. Machine learning is one of the most useful things when it comes to disease analysis because it is constantly improving, which allows it to make accurate predictions and classifications. This is one of the most useful aspects of machine learning[8]. If the predictive model is provided with additional data, it will be able to improve its decision-making capabilities, which will ultimately result in more accurate predictions. In order to extract useful insights from patient datasets that are stored in electronic medical records (EMRs), machine learning techniques can be utilized[9]. The data from electronic health records that are associated with a disease can be analyzed by machine learning algorithms, which can then make educated guesses as to what the actual causes of the illness are. Diagnostics of diseases can be significantly aided by this. When it comes to the analysis and integration of complex healthcare data sets, machine learning has surpassed the traditional bio statistical approach. This is especially true for tasks such as clustering, prediction, and classification [10]. The most recent developments in machine learning have enabled it to become exceptionally proficient in a wide range of tasks, such as the segmentation of medical images, the grouping of interstitial lung diseases, the reconstruction of medical images, and the identification of the components of brain tumors. There are a variety of fields that are beginning to recognize the growing significance of data-driven approaches and developments in machine learning[11]. Examples of supervised spam classifiers, advertising systems, fraud detection systems, and anomaly event detectors, respectively, that work to keep our email secure include systems that learn to match advertisements with context, systems that detect fraud, systems that protect banks from bad actors, and systems that assist experimental physicists in discovering events that lead to new theories in physics.. Two of the most important components that make it possible for these applications to function are the utilization of scalable learning systems to discover the relevant model from massive datasets[12] and the utilization of efficient statistical models to display complicated the relationships between data. The gradient tree boosting technique is a well-known machine learning method that has proven to be effective in a variety of settings. It has

demonstrated through experiments that tree boosting performs exceptionally well on a variety of classification algorithms that are widely utilized[13]. The LambdaMART algorithm is a result of modifications that were made to the tree boosting method.



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2. Related Works

The healthcare data, which is inherently diverse, is comprised of both structured and unstructured data. When it comes to effectively managing this diversity, it is not an easy feat to accomplish. There have been a lot of studies that have investigated different methods of data integration, such as multimodal learning and data fusion, in order to successfully handle the complexity of combining different types of data[14]. Electronic health records and high-throughput technologies make it possible to digitally record and analyze vast amounts of clinical and biomedical data. There is a wealth of information about intricate biomarker interaction systems contained within this data set. These systems open up exciting new avenues for the study of diseases. For the purpose of deriving and making this information possible, the data must be precisely structured, easily interpretable, and condensed. In the fields of statistics and machine learning, Bayesian networks (BNs) are utilized[15]. Bayesian networks are probabilistic models that are widely used and flexible in their application. They excel when it

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come to depicting intricate systems of interactions between different entities. Nodes in a Bayesian network (BN) are used to represent the features that have been measured, and directed edges are used to represent the connections that exist between these featured instances. The structure of this network makes it possible for BNs to depict multivariate interdependencies in an explicit manner[16]. As a result, they demonstrate how information is disseminated visually, which is something that can be comprehended by anyone. In contrast to other approaches to machine learning, they are distinguished by the fact that their structure can be interpreted. The discipline of Systems Biology makes extensive use of Bayesian network acquisition algorithms that have been demonstrated to be effective through automated processes. The applications of these algorithms that are most frequently used are cellular network simulation, modeling of protein signaling pathways, modeling of gene regulation networks, and modeling of medical decision support systems[17]. The computational inference of large biomedical datasets, on the other hand, is particularly challenging due to a number of obstacles. In addition, the dimensionality of these problems is quite high. The handling and structure of data, which includes operations such as annotation, normalization, discretization, imputation, and feature evaluation, requires a significant amount of human expertise in addition to a significant amount of physical labor[18]. Furthermore, these preprocessing procedures have a significant influence on the final model, in addition to the expenses that they incur. Bayesian network learning and hierarchical variable clustering are now components of an operational process that is entirely data-driven and fully automated. When it comes to manual processing, these issues are typically dealt with apart from one another; however, our methodology addresses multiple of them simultaneously. Utilizable biomedical data can be utilized to derive complex-controllable biomarker interaction models through the combination of these two widely accepted principles[19].

Bagging, boosting, and stacking are examples of ensemble methods that are utilized extensively in healthcare analytics. These methods are used to improve the accuracy of predictions. These techniques combine a number of different base learners, such as neural networks, decision trees, and support vector machines, in order to construct a prediction model



more trustworthy[20]. The use of healthcare ensemble learning has been demonstrated beneficial in a variety of domains, including disease prediction (for example, diabetes and cardiovascular diseases), measuring patient outcomes, and creating diagnostic support

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systems. When it comes to algorithms, XGBoost, Random Forests, and Gradient Boosting Machines (GBM) are among the most widely used. Ensemble learning is a technique that is widely utilized in the healthcare industry because it enhances the predictive accuracy of the system by combining multiple base models[21]. There are a variety of medical fields that have made use of the techniques of bagging, boosting, and stacking. The use of ensemble algorithmic techniques such as GBMs, Random Forests, and AdaBoost has proven to be successful in the prediction and diagnosis of diseases, particularly in the case of chronic diseases such as diabetes and heart disease. The meteoric rise in data collection can be attributed, in large part, to the proliferation of information-gathering devices[22]. These devices include air quality sensors, social media posts, digital photos and videos, records of purchases, and GPS signals from cell phones. The various findings related to this ensemble method is illustrated in table 1.

Table 1: Various Findings in Ensemble Method

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Author Nanni et.al.[23]		Proposes a heterogeneous ensemble using SVM, RB, and CNN.	Ensembles show promising performance on various medical datasets.	Requires careful tuning of parameters.
Mung et.al. [24]		Uses Naïve Bayes, K-NN, and decision tree for ensemble learning.	Ensemble classifiers outperform individual base learners.	May not be suitable for extremely large datasets.
Gupta et.al. [25]		Combines choice trees, random forests, and SVMs using AdaBoost.	Proposed ensemble method outperforms individual techniques.	Requires feature selection and balancing techniques.
Javale et.al. [26]		Uses KNN, SVM, Random Forest, Naïve Bayes, and Logistic Regression with stacking.	KNN and Random Forest perform well on balanced and imbalanced datasets.	May be computationally expensive for large datasets.

In the current body of literature on machine learning, a multitude of different approaches to ensemble construction are proposed. In order to generate multiple training datasets of the same size, it is necessary to replace some of the samples that were included in the initial dataset with random samples. Following the execution of a learning algorithm on every bootstrap sample, the classifiers that are produced are then averaged for the purpose of quantitative value prediction and combined through the utilization of a plurality vote for the purpose of class prediction[27]. Bagging does not have an effect on learning algorithms that are stable, such as k-nearest neighbor methods; however, it can significantly improve learning algorithms that are unstable, such as neural networks. Hybrid ensemble methods are used to improve the accuracy of predictions and to capture a wide variety of data patterns. These methods combine multiple ensemble techniques or base learners. When it comes to dealing



wide variety of healthcare data, such as imaging, electronic health records (EHRs), e genome data, hybrid ensembles frequently offer improved resilience, adaptability,

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and accuracy[28]. Personalized medical care, the diagnosis of chronic illnesses, and the detection of cancer are all examples of areas that make use of hybrid methodologies. The results of previous research have demonstrated that hybrid ensembles are capable of effectively managing healthcare data that is imbalanced, missing, and noisy.

2.1.Challenges in Healthcare Data Analytics

There is a strain being placed on healthcare data as a result of the growing prevalence of electronic health records (EHRs), wearable sensors, and genomic data. In order to keep up with this strain, hybrid ensemble methods need to be scalable. In the field of healthcare data, one of the most common problems is known as class imbalance[29]. This problem is characterized by the underrepresentation of certain outcomes, such as rare diseases. The Synthetic Minority Over-sampling Technique, also known as SMOTE, along with other hybrid ensemble methods are utilized in order to confront this problem. When deep learning models

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are incorporated into hybrid ensembles, the computational complexity of the ensembles in combination increases[30]. Researchers are currently concentrating their efforts on finding ways to enhance these models so that they can process healthcare data in real time to a greater degree of speed. The proliferation of a wide variety of medical data has been fueled in part by the proliferation of wearable devices and other healthcare institutions. Electronic health records, also known as EHRs, are the fundamental data storage system that underpins medical practice and makes it possible to extract insights from healthcare[31]. In the past, the majority of people who required medical attention would seek advice from more experienced physicians, while those who were dedicated to clinical research would carry out experiments that were both costly and meticulously planned. Nevertheless, the most important factor that is driving the implementation of data-driven methodologies is the current increase in access to electronic health records (EHR), which is opening the door to opportunities to automate actions that are associated with healthcare. There are a number of potential benefits, including more accurate prediction of outcomes, faster advancements in clinical research, improved suitability for patient care, and earlier identification of diseases[32]. While Big Healthcare Analytics is beginning to show signs of its potential, there is a significant gap between its potential and its actual application in the healthcare industry. The heterogeneity, timeliness, complexity, noise, and incompleteness of big data all work together to make it difficult to discover value from existing data. A consistent pattern is followed by Big Healthcare Analytics in the majority of instances. In order for an electronic health record (EHR) system to be utilized in an efficient manner, it is necessary to collect, integrate, clean, store, analyze, and comprehend all of the data that is contained within the system[33]. Every single thing is a data analysis pipeline, which is a process in which a number of different algorithms or systems are combined to produce a comprehensive solution, with each one aiming to achieve a particular objective. Furthermore, it is comparable to a software stack in which each stage provides a selection of different solutions. The selection of a solution is determined by the type of data being processed (for example, sensor data versus text data) or the requirements of the application (for example, predictive models versus cohort analysis). The use of hybrid ensembles that combine traditional



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learning with deep learning paradigms is becoming increasingly popular as a means of improving the extraction of features from complex healthcare data[34]. Examples of such data include medical images, genomic data, and time-series data from wearable devices. In

order to construct predictive models that do not compromise the confidentiality of patient information, we combine hybrid ensembles with federated learning, which is a technique that trains models by making use of decentralized data without information being shared.

3. Proposed Work

The data classification process refers to the arrangement of the available input data in accordance with the criteria that has been predefined. It is an important aspect in several applications. It helps to move forward with the other steps easily and with clarity.

3.1. Logistic regression algorithm

The probability of the occurrence of an event can be predicted through the logistic regression algorithm. It is widely used in the medical application fields. The features of the input data are mapped to an output variable whose values range between 0 and 1. Let us consider the input features and $i_1, i_2, i_3, \dots, i_n$ and the output variable as o . The range of values between 0 and 1 is the probability of occurrence of the event. The function of sigmoid is given by the following expression.

$$f(a) = \frac{1}{1 + e^{-a}} \quad (1)$$

Where the value of $a = u_0 + u_1 a_1 + u_2 a_2 + \dots + u_n a_n$. It represents the combination of the features at the input that are linear. The weights corresponding to these features are given by u . The optimized values of these weights can be determined by using the logistic regression algorithm. It helps to reduce the errors prevailing between the actual output and the predicted output. This can be attained by reduced the function of likelihood and it is given by the following equation.

$$L(u) = \prod_{j=1}^n f(a_j)^{y_j} (1 - f(a_j))^{1-y_j} \quad (2)$$

In the above equation n represents the number of samples involved in training. The actual output variable is given by y_i and the combination between the weights and the features at the inputs are given by a_i . The weights' optimized values can be determined by using the gradient descent. The gradient descent involves the updating of weights along the direction of negativity gradient in the likelihood function.

$$u_j := u_j - \beta \frac{\partial L(u)}{\partial u_j} \quad (3)$$

The likelihood functions partial derivative is given by $\frac{\partial L(u)}{\partial u_j}$. The rate of learning is given by β . The steps involved in the logistic regression algorithm are listed below.

1. Set the values of the weights $u_0, u_1, u_2, \dots, u_n$ values that are random.
2. The linear combination is determined for each of the training examples. This can be done by utilizing the current weights.
3. The value of the output that is predicted is determined for every example of training through the function of logistics.



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4. The error occurring between the prediction-based value and the actual output value is determined for every sample of training.
5. The likelihood function's gradient is evaluated in accordance with the respective weights.
6. The weights are updated using the gradient descent.
7. The steps two to six are iteratively done until the least error between the predicted value and the actual value are obtained.

The logistic regression algorithm is simple and easy to implement. It is one of those algorithms that is straight forward and statistically implemented by utilizing software. The probability of contributions done by each and every variable can be predicted through the logistic algorithm.

3.2.Detection of image

It is a process that involves placing the features, objects or parts of an image at a specific point in the image itself. It helps to gather more clarity of the image and helps in several applications like the finger print recognition, facial detection etc. In the proposed method, the image detection technique is used for the interpretation of different medical images.

3.3.YOLOv4 algorithm

YOLO stands for You Only Look Once. V4 represents that it is of version 4. The bounding boxes and the probabilities of the class can be predicted by using the YOLOv4 algorithm for every objects present within an image. For the extraction of features, this algorithm in turn used another technique called convolutional neural network.

For the prediction of the bounding boxes and the probabilities of the class, this algorithm splits the image into smaller segments called grids. The grids contain cells and this algorithm predicts the following aspects for every cell present in the grid.

1. The probability indicating whether that particular cell is present is that grid or not.
2. The bounding box's x and y coordinates which is relevant to the cell's coordinates.
3. The dimensions of the bounding boxes like its width and height. This is also related to the cells.
4. The probability of the class for each class of object.

The training of the YOLOv4 algorithm can be done by utilizing the loss function. The loss function is used to estimate the errors between the bounding boxes and the probabilities of the class. The mean square error is used for the calculation of this error.

3.4.Parallel advanced k means clustering and classification:

The parallel clustering technique has two important works. One is the management of the datasets and the other one is the performing of clustering operation. The performance of classification by the logistic algorithm can be improved with the help of the parallel advanced k means clustering. Figure 1 gives the architecture of parallel advanced k means clustering and classification.



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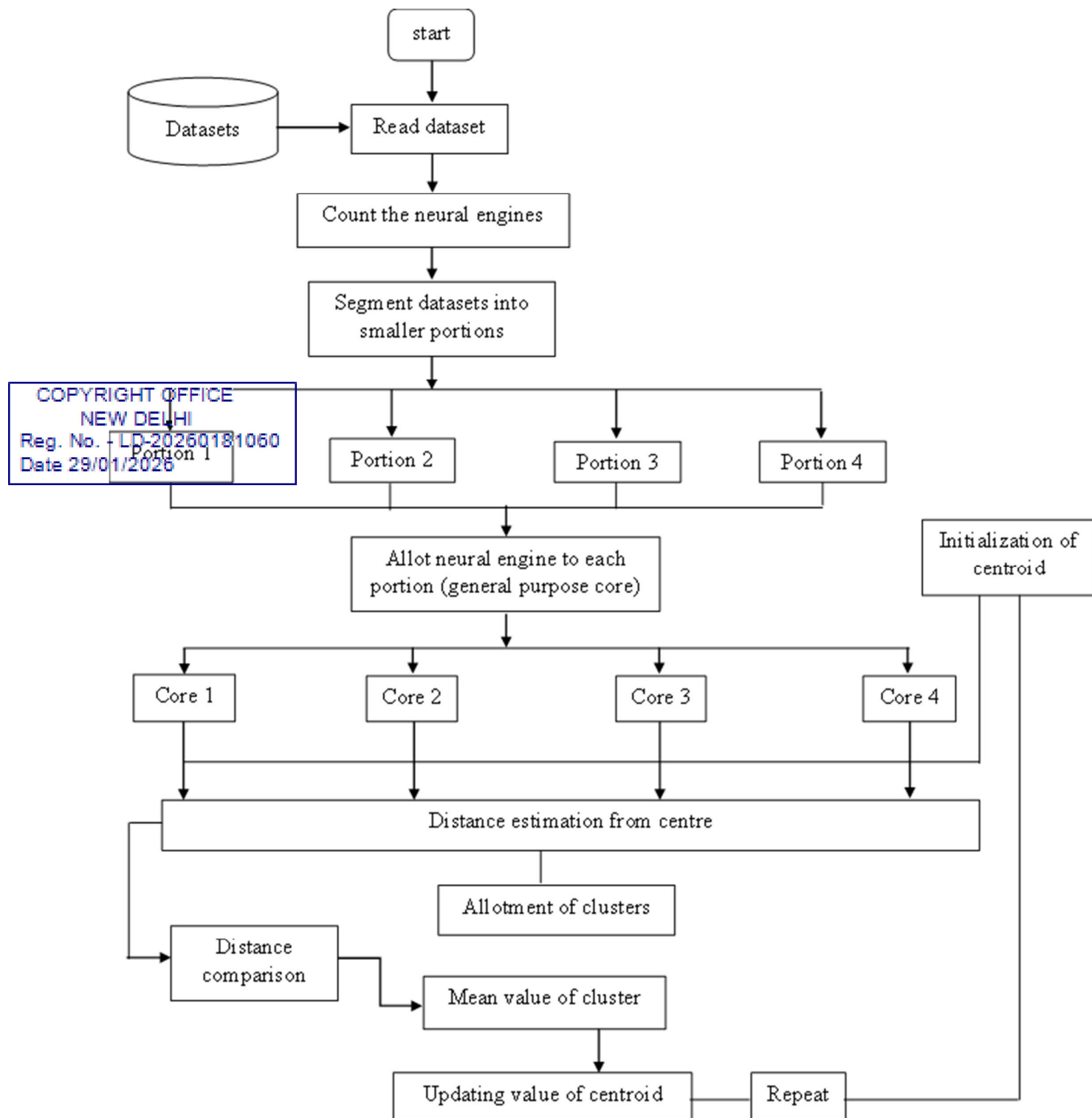


Figure 1: Architecture of parallel advanced k means clustering and classification



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Algorithm: Logistic regression parallel classification

Procedure: LOGISTIC REGRESSION PARALLEL CLASSIFICATION

Pre-processing of the dataset DS

Segment the dataset into two parts a) training (DS_{train}) b) testing (DS_{test})

Split the data into smaller portions called chunks DS_{train_1} , DS_{train_2} , DS_{train_n}

For $j \rightarrow 1$ to m do

Train the model for logistic regression on the training data chunk DS_{train_j}

End for

Associate the models $M_1, M_2, M_3, \dots, M_n$ to determine the eventual model M

Evaluate the final model M over the dataset of testing

Return the final model M

More data can be processed in an easier and in an efficient manner. The accuracy and the precision of the medical data can even be improved. The resources used for computation, the complexity involved in the classification can also be reduced.

3.3.1 Reprocessing and clustering of the input image:

As an initial step, the segmentation of the X ray and the Ct scan images were done using the k means clustering approach. The result of this step would be image that is split into smaller segments that can be used for further analysis.

The input images were converted to have a size of 620 x 620 pixels through the normalization procedure. The training data was matched with the YOLOv4 model by utilizing the proposed k means clustering algorithm.

Algorithm: K means clustering

Requirement: Datasets containing the medical images

Inputs: Random point of centroid

Begin grouping of pixels (formation of clusters)

While pixels not equal to end do

Choose the neural engine core for each of the group/cluster

Allot the core for processing

Determine the value of the mean

Assign the pixels to clusters

End while

Output: grouped pixels

By using the clustering technique, the colour and tone variations present in the image can be reduced, which in turn helps in the removal of the unwanted noises in the image. After the clustering of the pixels, the pixels will be assigned to different clusters. Because of the smaller number of colours and tones, the clustered image can be used for the extraction of features with less complexity. The medical images like the x ray images and the CT scan images were segmented based in the presence of tissues, organs and bones. These images were further evaluated for the extraction of features. The selection of features depends on the size, shape



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3.6.Detection of image by using YOLOv4 algorithm

The objects in the images can be handled at a high speed by the scanning carried by the YOLOv4 algorithm. Since in the prior step, the images have been segmented by the formation of clusters, the identification of the objects in the image becomes easier in this step. The image is compressed and processed by the utilization of the convolutional neural network. Predictions can be made at the end of the classification process. Though this method is efficient, it has its own limitation. It does not have the capability to identify the objects that are smaller in a precise manner. The input images were segment into grids of sizes $S \times S$. each grid is responsible for identifying the centre of the object in case if the cell lies within that grid. The core of the confidence is defined by the following equation.

$$\text{Confidence Score} = P(\text{class} | \text{object}) \times P(\text{object}) \quad P(\text{object}) \in [0,1] \quad (4)$$

$P(\text{object})$ represents the probability of an object present in the grid. $P(\text{class}|\text{object})$ indicates that the probability of presence of an object in a class depending upon the availability of an

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3.7.YOLOv4 – K means clustering

The performance of the proposed system can be increased by reducing the inference time delays. The evaluation of the algorithm can be done through the functions of activation.

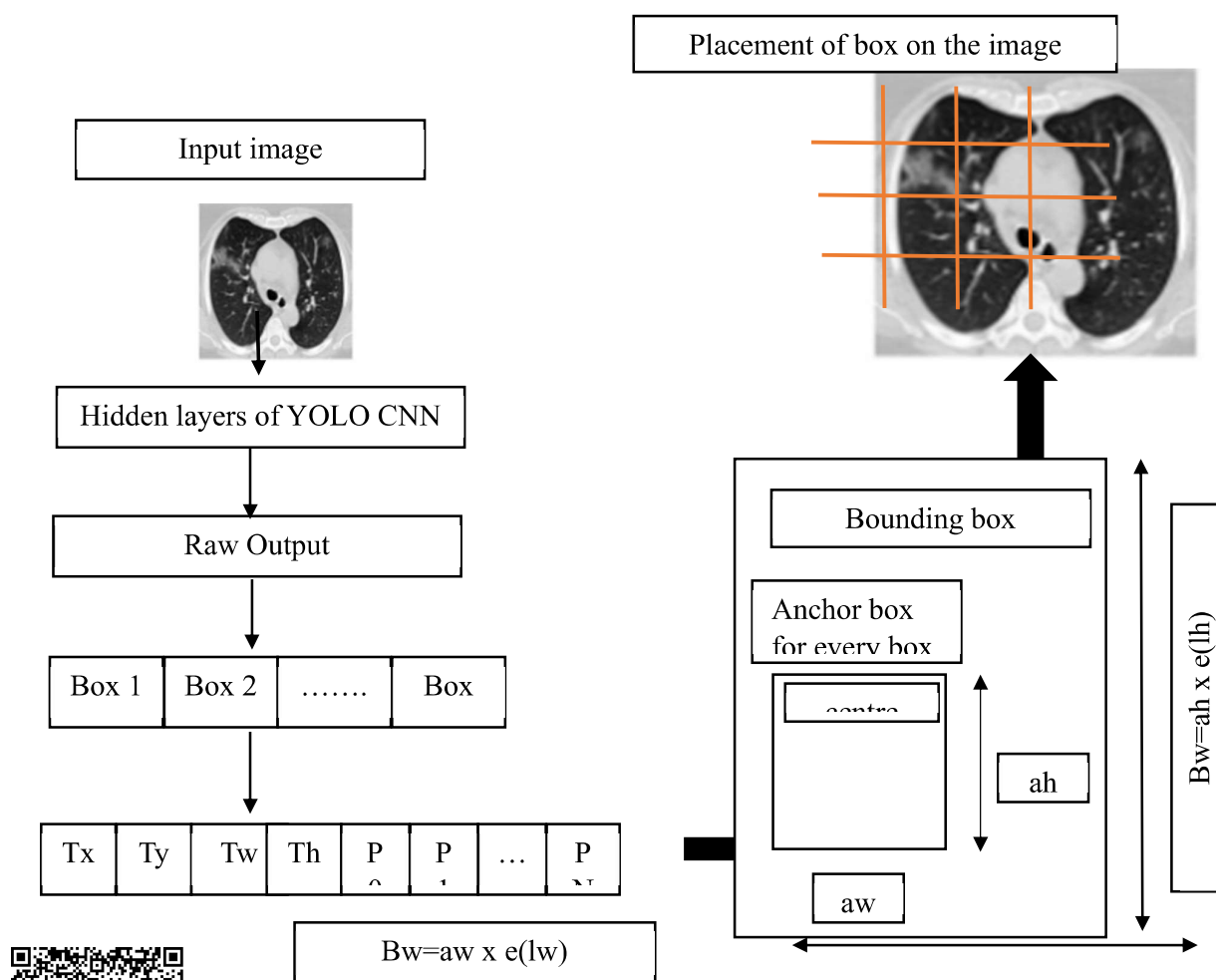


Figure 2: Architecture of image detection through YOLOv4 algorithm



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The usually used algorithms like the ReLU are not efficient enough to promote creation of features and this led to the development of a new technique that has been proposed over here.

Algorithm: Hybrid K means YOLOv4 classifier

Requirement: Datasets containing the medical images

Inputs: Random point of centroid

Begin grouping of pixels (formation of clusters)

While pixels not equal to end do

Choose the neural engine core for each of the group/cluster

Allot the core for processing

Determine the value of the mean

Assign the pixels to clusters

End while

Run the YOLOv4 model on the clustered image

If image has COVID then

Set the flag as “affected image”

Else

Set the flag as “non affected image”

End if

Output: classified image

Figure 2 give the architecture of image detection through YOLOv4 algorithm. The clustered image is received by the YOLOv4 detector. This takes place before the initialization of the layers of YOLOv4 over the image. The performance of the layers can be improved by the clusters present in the image. This helps in improving the detection of objects, features and other stuffs in the image. The input image is fed as an input to the advanced K means parallel clustering algorithm. This algorithm along with the combination of the YOLOv4 classification technique produces the clustered image pixels. This is obtained at the output.

3.8.XG boost algorithm for image enhancement

The XG boost algorithm is a combination of mixed sampling and ensemble learning approaches. It is capable of training several decision trees and produce a combined result. It is more advantageous when the number of features under consideration are more. In the proposed approach, it has been used for the enhancement and optimization of the resultant image obtained through the hybrid K means YOLOv5 method.

Let us consider the training set to have m number of examples and n number of features.

$d = \{(y_j, x_j)^m_{j=1} \mid d \models m, y_j \in R^n, y_i \in R\}$. This XG boost algorithm is a model that is of

additive in implementation. The predicted value is represented by $g_k(y_j), g_k \in G$. The value that is predicted is obtained after providing the ith sample as the input. The XG boost algorithm definition is given by the following equation.

$$Object = \sum_{j=1}^m l(x_i, x_i) + \sum_{k=1}^K \delta(g_k) \quad (5)$$



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The real result is represented by x_j and the loss function is denoted by $\sum_{k=1}^K \delta(g_k)$. The

original model will remain unchanged during the application of XG boost algorithm. The next set of trees will be set as residues.

The steps involved in the XG boost algorithm are given below.

$$\hat{x}^{(1)} = g_1(y_j) = \hat{x}^{(0)} + g_1(y_j) \quad (6)$$

$$\hat{x}^{(2)} = g_1(y_j) + g_2(y_j) + g_{21}(y_j) \quad (7)$$

$$\hat{x}^{(t)} = \sum_{k=1}^t g_k(y_k) = \hat{x}^{(t-1)} + g_t(y_j) \quad (8)$$

In the above equations, $x_j(t)$ indicates the value that is predicted by the model during the current iteration and $x_j(t-1)$ indicates the value that is predicted in the previous iteration.

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$$Object^{(t)} = \sum_{j=1}^m h_j g_t(y_j) + \frac{1}{2} h_j g_t^2(y_j) + \rho(g_t) \quad (9)$$

$$Object^{(t)} = \sum_{j=1}^m h(x_j, \hat{x}_j^{(t)}) + \sum_{l=1}^{t-1} \rho(g_l) + \rho(g_t) \quad (10)$$

$$Object^{(t)} = \sum_{j=1}^T h_j g_t(y_j) + \frac{1}{2} h_j g_t^2(y_j) + \lambda T + \frac{1}{2} \gamma \sum_{i=1}^T v_i^2 \quad (11)$$

$$= \sum_{i=1}^T \left(\sum_{j \in J} f_j \right) v_i + \frac{1}{2} \left(\sum_{j \in J} h_j + \gamma \right) v_i^2 \quad (12)$$

$X_j(t)$ and $l(x_j, x_j(t-1))$ are constants. Hence the object function can be estimated by using the Taylor expansion of the loss function. The second order Taylor expansion is used for the approximation of the objective function. Leaving out the values that are constants, we will have the following equation.

$$Object^{(t)} = \left[\sum_{j=1}^T G_i v_i + \frac{1}{2} (H_i + \gamma) v_i^2 \right] + \lambda T \quad (13)$$

With an assumption that the structure of the tree has been identified, the optimized solution determined through derivation is given by

$$Object^{(t)} = -\frac{1}{2} \sum_{i=1}^T \left(\frac{G_i^2}{H_i + \gamma} \right) + \lambda T \quad (14)$$

The gain value of the XG boost algorithm is calculated using the following formula. It makes the selection of the feature that has the highest value of gain. The point of split is chosen based on this value of the gain.

$$Gain \text{ Value} = \frac{1}{2} \left(\frac{G_l^2}{H_l + \gamma} + \frac{G_r^2}{H_r + \gamma} - \frac{(G_l + G_r)^2}{H_l + H_r + \gamma} \right) - \gamma \quad (15)$$

$$Gain \text{ Value}' = \frac{1}{2} \left(\frac{(G_l - \gamma)^2}{H_l + \gamma} + \frac{(G_r - \gamma)^2}{H_r + \gamma} - \frac{(G_l + G_r - \gamma)^2}{H_l + H_r + \gamma} \right) - \frac{\gamma}{2} \quad (16)$$



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The core structures of the left tree and the right tree are represented by $\frac{G_l^2}{H_l + \gamma}$ and $\frac{G_r^2}{H_r + \gamma}$ respectively. When the splitting of the current node has not been accomplished, the score of the structure during this time is given by $\frac{(G_l + G_r)^2}{H_l + H_r + \gamma}$. The over fitting problem can be avoided and a minimized variance of the model can be obtained.

4. Results And Discussion

Table 2 gives the grouping of datasets. For the COVID 19 dataset with the size of 2GB, the count of the image is 7594 and the class count is about 10. The class count and the image count for the Slice CT scan COVIDx CT are 11 and 194933 respectively. The size of this dataset is about 65GB. The values of the class count and the image count are the highest for the Slice CT scan COVIDx CT datasets among the others considered. The Low dose CT Reconstruction a size of 20GB has image count of 16928 and class count of 7.

Table 2: Grouping of datasets (Clustering)

S.No	Datasets	Size of the data (GB)	Image counts	Class count
1.	COVID 19	2	7594	10
2.	Slice CT scan COVIDx CT	65	194933	11
3.	Low dose CT Reconstruction	20	16928	7

Table 3 gives performance measures of the proposed algorithm with different data sets. When the number of folds is one, the recall, precision, F1 scores and accuracy of the X ray datasets are observed to be 83.30, 83.61, 83.72 and 83.24 respectively. For the fold count with one, the CT dataset has produced the highest recall value and precision. The highest F1 score is produced by the X ray datasets and the highest accuracy is obtained in the MRI datasets. Recall, precision, F1 scores and accuracy of the CT datasets are observed to be 84.52, 84.56, 80.12 and 80.25 respectively and the same measures in case of the MRI datasets are observed to be 83.52, 83.35, 83.65 and 83.64 respectively

Table 3: Performance measures of the proposed algorithm with different data sets

E	Datasets	Recall	Precision	F1 score	Accuracy
1	X ray	83.30	83.61	83.72	83.24
	CT	84.52	84.56	80.12	80.25
	MRI	83.52	83.35	83.65	83.64
2	X ray	94.41	94.41	94.35	94.26
	CT	94.56	95.45	93.64	93.87
	MRI	94.56	94.82	94.67	94.48
3	X ray	94.30	95.24	94.62	95.36
	CT	94.35	95.62	90.26	90.68
	MRI	95.68	95.48	94.84	94.72



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4	X ray	97.81	97.41	98.35	97.85
	CT	98.35	98.68	97.51	97.68
	MRI	97.25	97.82	97.62	97.13
5	X ray	98.21	98.62	98.43	98.68
	CT	99.53	99.64	99.68	99.62
	MRI	98.52	98.71	98.32	98.24

When the fold count is 2, the minimum recall value is obtained by using the X ray dataset. Both the CT and MRI datasets have the same recall values of 94.56 whereas it is 94.41 in case of the X ray datasets. The maximum precision is obtained through the CT dataset with a value of 94.82. the precision of other two datasets are also almost the same. It is 94.1 and 94.82 in case of X ray and MRI respectively. The F1 score of the X ray, CT and MRI are 94.35, 93.64 and 94.67 respectively and their accuracy values are 94.26, 93.87 and 94.48 respectively.

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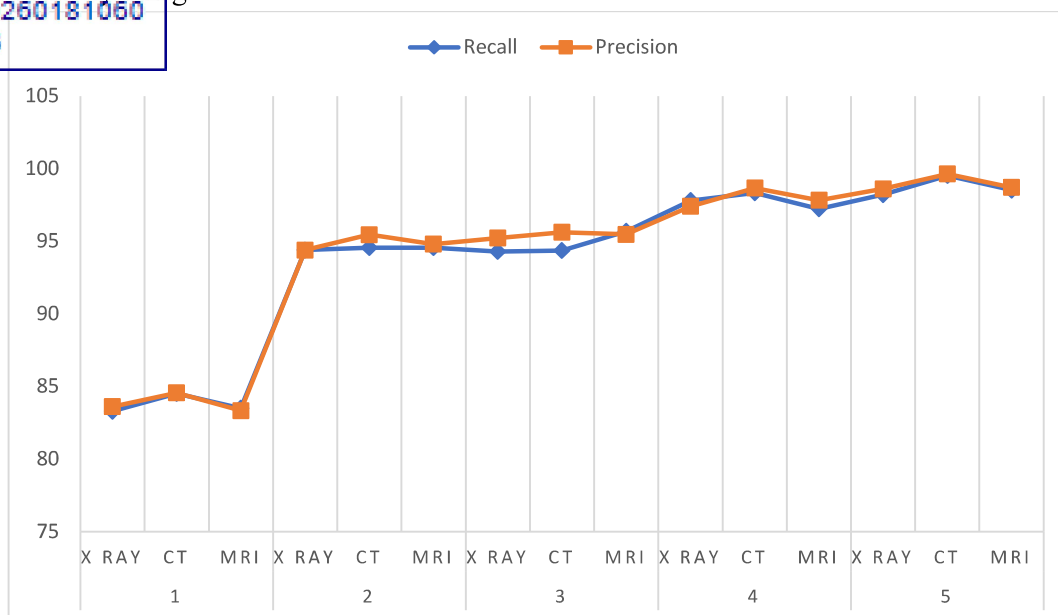


Figure 3: Recall and precision of the X ray, MRI and CT datasets

When the number of folds is three, the recall, precision, F1 scores and accuracy of the X ray datasets are observed to be 94.30, 95.24, 94.62 and 95.36 respectively. For the fold count with three, the MRI dataset has produced the highest recall value. The highest F1 score is also produced by the MRI datasets and the highest accuracy is obtained in the X ray datasets. Recall, precision, F1 scores and accuracy of the CT datasets are observed to be 95.68, 95.48, 94.84 and 94.72 respectively and the same measures in case of the MRI datasets are observed to be 95.68, 95.48, 94.84 and 94.72 respectively.



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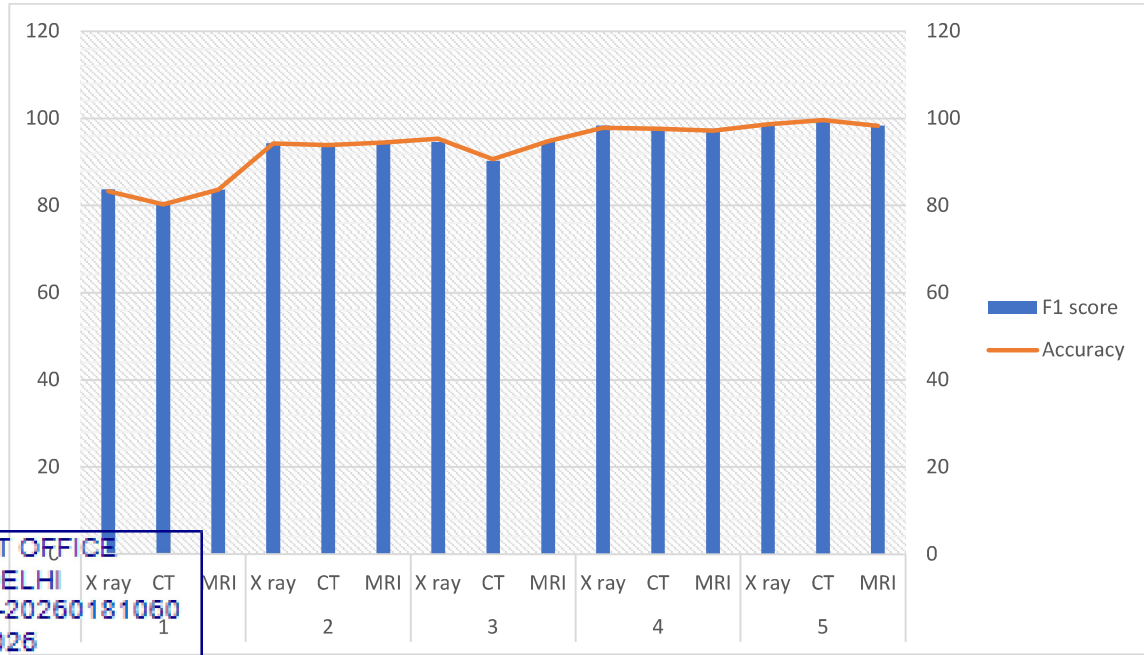


Figure 4: Accuracy and F1 score of the X-ray, MRI and CT datasets

Figure 3 depicts the recall and precision of the X-ray, MRI and CT datasets and figure 4 represents the accuracy and F1 score of the X-ray, MRI and CT datasets. When the fold count is four, the minimum recall value is obtained by using the MRI dataset whose value is about 97.25. The values of the other two, namely the CT and X ray are 98.35 and 97.81 respectively. The maximum precision is obtained through the CT dataset with a value of 98.68. The precision of other two datasets are also almost the same. It is 97.41 and 97.82 in case of X ray and MRI respectively. The F1 score of the X ray, CT and MRI are 98.35, 97.51 and 97.62 respectively and their accuracy values are 97.85, 97.68 and 97.13 respectively. The value of F1 score and accuracy produced by the CT dataset are the highest.

Table 4: Object detection speed of various algorithms

S.No	Algorithm	Speed in ms (time taken to detect objects)
1	SPP-net	1600
2	R-CNN	1000
3	Fast R-CNN	850
4	Faster R-CNN	700
5	R-FCN	650
6	Mask R-CNN	500
7	YOLOv3	350
8	YOLOv5	250
9	KAPC-YOLOv4	190



Table 4 gives the speed of object detection with the several algorithms. A comparison of speed values in milli seconds has been done with the proposed algorithm to prove how fast the proposed model is. The object can be detected in a shorty period of time by using

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the proposed algorithm. It only takes about 190ms. The other methods have higher object detection times when compared to the proposed model. The time taken to detect the object by the SPP net and recurrent convolutional (R-CNN) are 1600 milli seconds and 1000 milli seconds respectively. Fast RCNN takes less time when compared to the convenient RCNN.

The time it takes is 850 milli seconds which is still lesser than the proposed model. The faster RCNN taken even more less time than the RCNN and the fast RCNN. It detects the objects in the image in about 700 milli seconds. For the detection of objects, R-FCN takes 650 milli seconds, mask RCNN takes 500 milli seconds, YOLOv3 takes 350 milli seconds and YOLOv5 takes 250 milli seconds. It can be clearly seen that the proposed hybrid model with K means advanced parallel clustering and YOLOv4 has taken the least time for the detect of object images and it is therefore more efficient in terms of the speed.

Table 5: Parameter comparison of various tree boosting algorithms

Technique	Greedy exact	Global approx	Local approx	Out of the core	Sparsity awareness	simultaneous
Boost	✓	✓	✓	✓	✓	✓
pGBRT	X	X	✓	X	X	✓
Spark MLib	X	✓	X	X	Partial	✓
H2O	X	✓	X	X	Partial	✓
Scikit learn	✓	X	X	X	X	X
R GBM	✓	X	X	X	Partial	X

Table 5 gives the parameter comparison of various tree boosting algorithms. The comparison is done based on the presence and the absence of the parameters namely, greedy exactness, global approximation, local approximation, out of the core, sparsity awareness and simultaneous processing. The pGBRT algorithm has local approximation but does not have global approximation. The greedy exactness is absent. It can perform simultaneous operations but it lacks the sparsity awareness.

Table 6: Data classification speed and accuracy of the classification algorithms

Dataset	Algorithm	Speed in ms	Accuracy in %
1	Logistic regression	11.2	94.5
	Naïve bayes	7.6	93.2
	K means – logistic regression	8.6	96.1
2	Logistic regression	62.1	95.3
	Naïve bayes	92.2	94.6
	K means – logistic regression	44.3	98.5
3	Logistic regression	27.45	94.2
	Naïve bayes	35.12	92.5
	K means – logistic regression	16.78	98.6



The Spark MLib has global approximation and can perform simultaneous operations. It does not have exact greediness and cannot perform local approximations. The sparsity

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awareness is partially present. Both H2O and Scikit learn lack performance of local approximations. The sparsity awareness is partially available with H2O and completely absent with Scikit learn. The simultaneous processing can be done by the H2O and cannot be carried out by the scikit learn. Scikit learn has only greedy exactness. The identities of the R GBM are almost same as that of the scikit learn except in case of the sparsity awareness. It is partially present in case of the R GBM boosting algorithm. Looking at the boosting algorithm used in the proposed methodology, which is the XG boost algorithm, it holds all the measures completely. IT does not have lacking or partial measures of any of the performance characteristics.

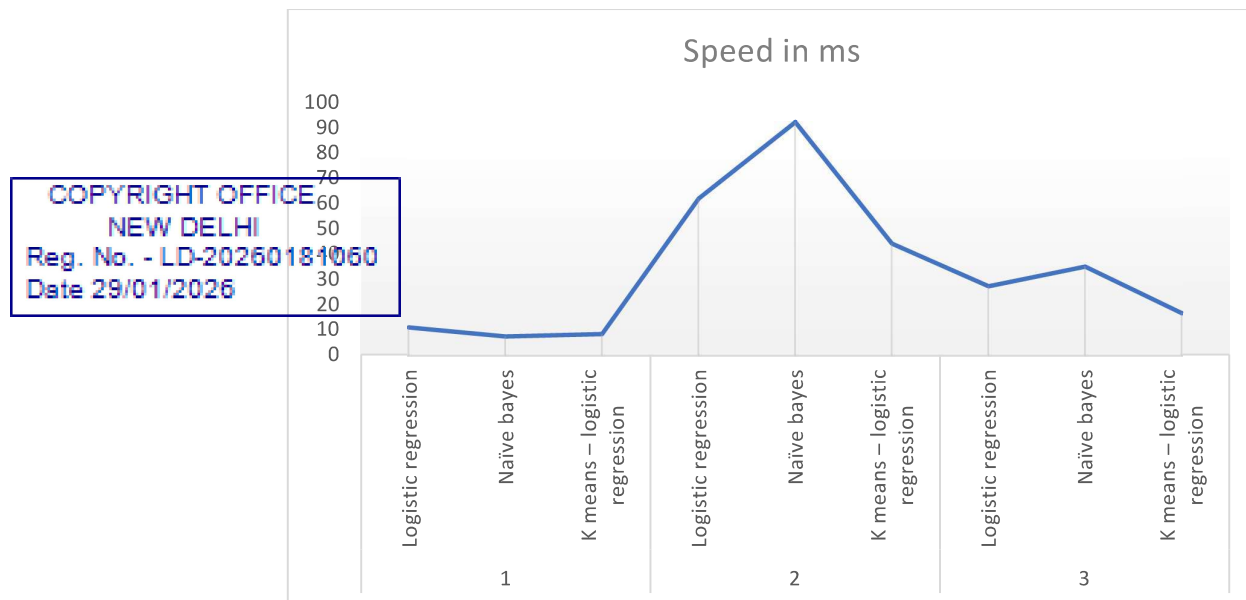
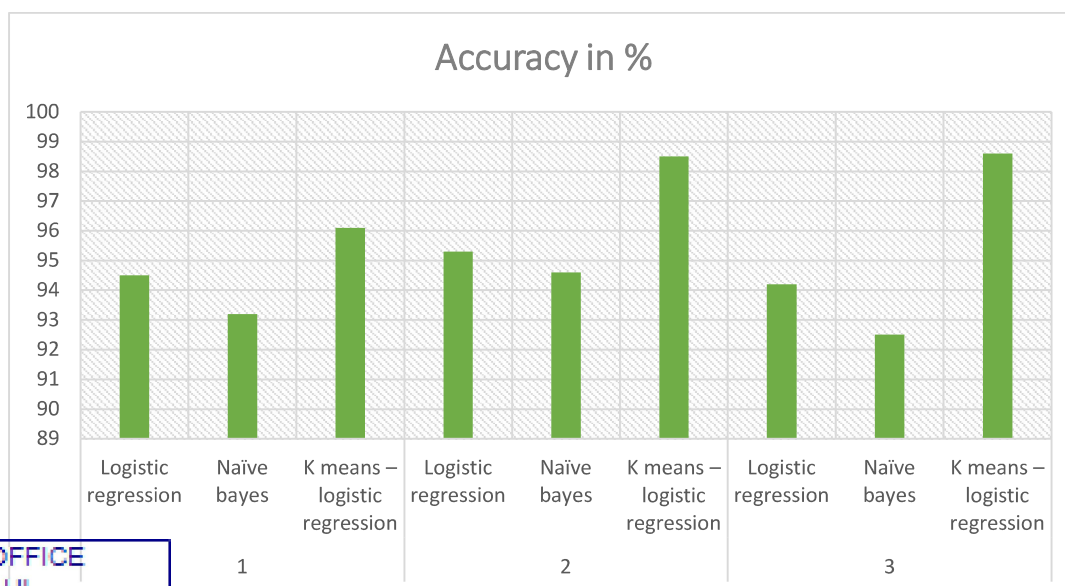


Figure 5: speed of classification algorithms in milli seconds.

Table 6 lists the speed and accuracy produced through the classification algorithm. The classification algorithm used in the proposed model is K means logistic regression. Its speed and accuracy in classification are compared with other techniques like the traditional logistic regression and naïve bayes for three different datasets. For the first type of datasets, the speed in milli seconds of the Logistic regression, Naïve bayes and K means – logistic regression is 11.2, 7.6 and 8.6 respectively and their percentages of accuracies are 94.5, 93.2 and 96.1 respectively. For the second type of datasets, the speed in milli seconds of the Logistic regression, Naïve bayes and K means – logistic regression is 62.1, 92.2 and 44.3 respectively and their percentages of accuracies are 95.3, 94.6 and 98.5 respectively. For the third type of datasets, the speed in milli seconds of the Logistic regression, Naïve bayes and K means – logistic regression is 27.45, 35.12 and 16.78 respectively and their percentages of accuracies are 94.2, 92.5 and 98.6 respectively. Figure 5 depicts the speed of classification algorithms in milli seconds. Figure 6 represents the accuracy of the classification algorithms in percentage.



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Figure 6: Accuracy of the classification algorithms in percentage.

Table 7 give the time taken per tree and area under the curve of 1000 trees on Higgs 1M data. The time per each tree using XG boost algorithm is 0.6942 and by using Higgs Bason, it is 0.6541. Scikit learn and R GBM takes 0.2962 and 1.0752 seconds respectively. The area under the curve of XG Boost and Higgs Bason are 0.8305 and 0.8264 respectively. The area under the curve of Scikit learn and R GBM are 0.8352 and 0.6248 respectively.

Table 7: Time taken per tree and area under the curve of 1000 trees on Higgs 1M data

S.No	technique	Time per each tree in seconds	Area under the curve
1.	XG Boost	0.6942	0.8305
2.	Higgs Bason	0.6541	0.8264
3.	Scikit learn	0.2962	0.8352
4.	R GBM	1.0752	0.6248

Table 8 gives the time taken for execution by serial and parallel k means clustering. The time difference between the two types of clustering techniques have been estimated and the efficiency has been identified with the percentage of speed observed. Figure 7 gives the graphical representation of execution times of serial and parallel k means clustering approaches and figure 8 depicts the graphical curve of the percentage of speed up in parallel k means clustering when compared to the serial k means clustering.

Table 8: Time taken for execution by serial and parallel k means clustering

Count of datasets	Time taken for execution (sec)		Difference (Ts-Tp)	Percentage of speed up
	Serial K means clustering (Ts)	parallel k means clustering (Tp)		
1	0.032	0.031	0.001	3.24
2	8.46	5.04	3.42	40.48



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100500	16.60	10.19	6.41	38.61
200500	34.04	21.93	12.11	36.59
300500	66.07	50.93	15.14	22.96
400500	82.35	63.04	19.31	24.55
600500	94.68	69.36	25.32	27.74
700500	113.07	90.19	22.88	21.24
800500	135.21	101.19	34.02	25.16
900500	173.68	124.80	48.88	28.15

When the number of datasets is 500, the serial k means clustering takes 0.032 seconds for execution and parallel k means clustering takes 0.031 seconds for execution. The time taken by the serial and parallel techniques in case of 50500 datasets are 8.46 seconds and 5.04 seconds respectively. The serial k means clustering takes 34.04 seconds, 66.07 seconds and 82.35 seconds for the dataset numbers of 200500, 300500 and 400500 respectively. The parallel k means clustering takes 21.93 seconds, 50.93 seconds and 63.04 seconds for the dataset numbers of 200500, 300500 and 400500 respectively.

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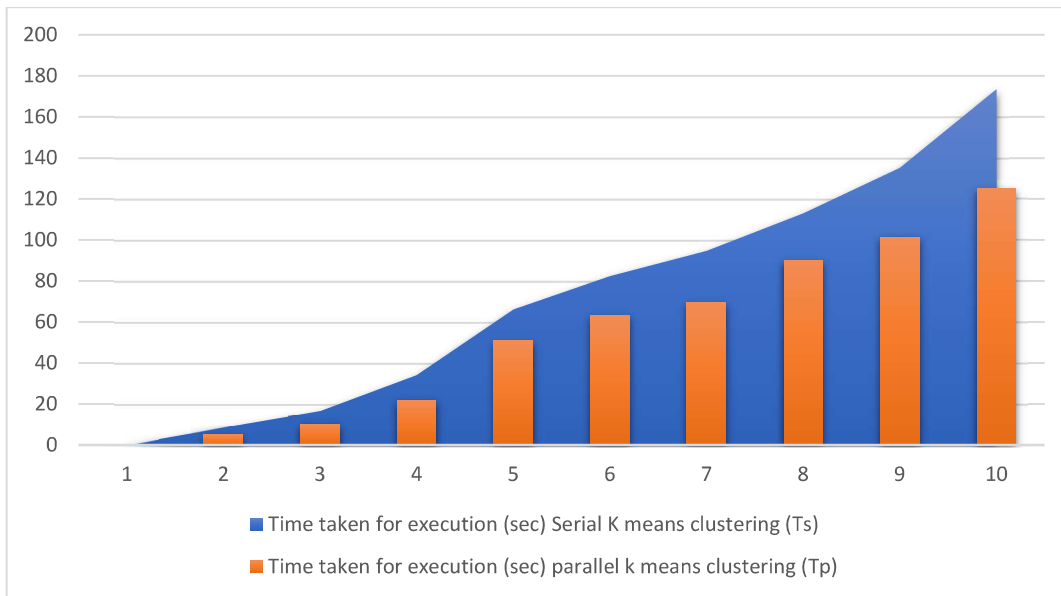


Figure 7: Execution times of serial and parallel k means clustering approaches

About 600500 datasets are executed in 94.68 seconds by the serial method of clustering and in about 69.36 seconds in case of the parallel clustering method. The execution times of the serial k means clustering technique for the dataset counts of 700500, 800500 and 900500 are 113.07 seconds, 135.21 seconds and 173.68 seconds respectively. Similarly, the execution times of the serial k means clustering technique for the dataset counts of 700500, 800500 and 900500 are 90.19 seconds, 101.19 seconds and 124.80 seconds respectively.



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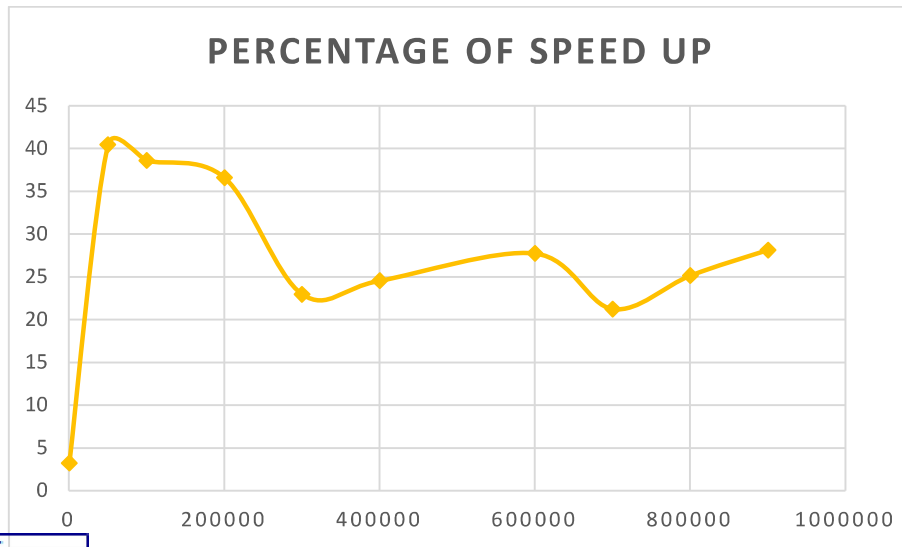


Figure 8: Percentage of speed up in parallel k means clustering when compared to the serial k means clustering

The execution times in case both the clustering techniques increase as the number of datasets increases. But it can be observed that under each circumstance, the execution time taken by the parallel clustering method is lesser than that of the serial clustering method. The difference in the time periods obtained for every dataset count considered are also determined and included in the tabulation for identifying the exact reason for choosing the parallel means of clustering technique in the proposed model. The percentage of speed up indicates the speed efficient of the clustering techniques with the increase in the number of datasets.

5. Conclusion

In the proposed method, hybrid machine learning algorithm has been proposed for the analysis of the medical data. K means advanced parallel clustering algorithm has been employed to preprocess the image which involves the removal of the unwanted noise from the raw input image. The classification and detection of objects in the image has been carried out through logistic regression algorithm and YOLOv4 algorithms respectively. The enhancement of the image has been done at the end using the XG boost algorithm. The performance parameters considered for the evaluation of the results are precision, recall, accuracy and F1 score. A comparison has been made between the speeds of detecting the objects among the proposed algorithm and the existing algorithm. In all the analysis done, the proposed algorithm has been observed to produce the best results.



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